

CARTWRIGHT, NL, Canada

WMO: 718180

Lat: 53.683N

Lon: 57.041W

Elev: 47

StdP: 14.67

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 2	(b) -15.4	(c) -11.6	(d) -22.7	(e) 1.6	(f) -14.5	(g) -18.7	(h) 2.0	(i) -10.9	(j) 43.4	(k) 22.2	(l) 39.8	(m) 20.2	(n) 11.8	(o) 210	(p) 0.843

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 16.4	(c) 76.6	(d) 63.3	(e) 72.5	(f) 61.1	(g) 68.9	(h) 59.5	(i) 64.9	(j) 73.4	(k) 62.7	(l) 70.5	(m) 60.6	(n) 67.3	(o) 10.3	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	81.4	68.7	59.2	75.3	66.9	57.0	69.5	64.8	30.0	73.5	28.4	70.4	26.9	67.3	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
35.5	29.5	25.1	-20.3	85.5	3.0	2.6	-22.5	87.4	-24.2	88.9	-25.9	90.3	-28.0	92.2
			-20.7	71.3	2.9	3.4	-22.8	73.8	-24.5	75.7	-26.1	77.6	-28.2	80.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	9.0	8.7	17.3	29.2	38.3	48.5	55.4	56.0	48.9	39.7	29.7	18.4		
	DBStd	18.59	12.39	11.70	10.76	8.08	6.44	8.27	7.33	6.48	6.48	5.78	7.16	10.63		
	HDD50	6624	1271	1156	1015	624	370	129	30	17	99	323	610	980		
	HDD65	11560	1736	1576	1480	1074	827	499	306	289	485	783	1060	1445		
	CDD50	561	0	0	0	0	8	83	199	202	64	5	0	0		
	CDD65	22	0	0	0	0	0	3	9	9	1	0	0	0		
	CDH74	242	0	0	0	0	3	47	105	80	7	0	0	0		
	CDH80	39	0	0	0	0	1	8	16	13	1	0	0	0		
(14) Wind	WSAvg	11.8	13.6	13.4	13.5	12.2	10.4	10.2	8.6	8.4	11.1	12.2	13.8	14.5		
(15) Precipitation	PrecAvg	41.7	3.6	3.0	4.2	3.3	2.7	3.7	3.6	3.4	3.5	3.3	3.8	3.7		
	PrecMax	52.2	9.5	10.1	11.4	8.5	5.7	7.4	6.9	5.1	8.7	6.2	7.6	7.2		
	PrecMin	31.2	0.9	0.6	1.1	0.2	0.9	1.3	1.5	1.3	1.7	0.9	1.7	1.1		
	PrecStd	6.5	2.0	2.1	2.5	2.0	1.1	1.4	1.3	1.1	1.5	1.5	1.6	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	40.0	43.1	53.9	67.8	80.1	82.2	81.7	73.2	61.1	52.8	42.3		
		MCWB	37.0	37.8	38.1	45.3	54.4	64.1	65.4	66.8	62.1	53.6	50.1	40.8		
	2%	DB	34.2	34.7	38.4	47.8	60.0	73.0	76.8	75.4	68.0	55.3	45.9	37.6		
		MCWB	33.3	33.9	35.1	41.1	49.9	60.0	63.6	63.2	59.4	50.4	43.3	36.6		
	5%	DB	31.9	32.1	34.7	43.5	54.1	68.3	72.6	71.2	63.8	51.6	41.5	34.8		
		MCWB	31.5	31.6	33.1	38.7	46.3	57.3	61.3	61.4	56.9	47.4	39.3	33.9		
	10%	DB	29.0	28.1	32.6	39.9	49.1	62.9	68.6	67.3	59.7	48.3	38.6	32.8		
		MCWB	28.1	27.2	31.5	36.3	43.2	54.8	59.8	59.6	53.6	44.8	36.9	32.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	38.0	39.4	46.1	56.0	65.5	68.1	67.9	64.2	54.8	50.4	40.7		
		MCDB	38.4	39.4	42.1	52.3	65.1	76.7	77.9	78.0	71.1	58.5	52.6	41.8		
	2%	WB	33.5	34.1	35.5	41.9	50.6	61.5	64.9	64.8	60.7	50.9	43.6	36.6		
		MCDB	34.2	34.5	37.6	46.8	58.9	70.5	73.5	72.7	66.5	54.6	45.7	37.4		
	5%	WB	31.5	31.7	33.2	39.0	46.6	58.3	62.7	62.6	57.3	47.8	39.8	34.0		
		MCDB	32.0	32.2	34.5	43.1	53.8	66.6	70.7	69.5	62.6	51.0	41.0	34.5		
	10%	WB	28.3	27.3	31.6	36.5	43.2	55.2	60.5	60.2	54.1	45.0	37.1	32.2		
		MCDB	29.1	28.3	32.9	39.5	48.6	62.8	67.3	66.5	58.9	48.0	38.4	32.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	15.7	15.5	13.2	13.5	16.7	17.6	16.4	13.7	9.5	8.9	10.9		
		MCDBR	15.0	15.6	16.8	19.1	25.7	27.5	25.7	24.1	20.9	15.9	12.1	10.6		
		MCWBR	14.9	15.4	14.8	13.6	16.5	16.5	14.9	14.1	13.8	12.0	11.2	10.5		
	5% WB	MCDBR	14.4	15.4	15.3	17.0	24.1	26.4	23.1	22.3	20.0	15.3	11.7	10.0		
		MCWBR	14.4	15.3	13.7	12.5	16.1	17.1	15.0	14.3	14.6	12.3	11.2	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.277	0.296	0.338	0.370	0.376	0.405	0.370	0.342	0.315	0.279	0.261		
	taud		2.273	2.249	2.218	2.168	2.313	2.393	2.357	2.455	2.505	2.540	2.458	2.263		
	Ebn at Noon		231	262	281	281	278	277	266	270	265	250	224	201		
	Edh at Noon		18	26	34	41	38	36	36	31	26	20	16	16		
(44) All-Sky Solar Radiation	RadAvg		255	510	879	1341	1505	1551	1499	1270	932	504	261	174		
	RadStd		42	76	165	137	184	157	105	94	61	40	23	22		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.70	N/A	N/A	N/A	N/A

Nomenclature: See separate page